

There exists a phenomenal growth opportunity in the voice and music arena

Designing audio devices used to be simple, but modern designs are very integrated and efficient. This interview looks at what has changed, and helps you keep on top of the changes. Joep A.J. van Beurden, chief executive officer, CSR plc speaks with Dilin Anand, senior assistant editor at EFY

JOEP A.J. VAN BEURDEN
CHIEF EXECUTIVE OFFICER, CSR PLC



What is transforming the audio electronics industry?

What we see now is a single chip, a piece of silicon that has audio codec, memory, processor and power management all rolled into one. This is like a complete voice and music platform. For instance, if you open a pair of headphones, there is usually just one chip inside. This single chip takes care of the entire sound quality and music experience that you have when you listen to it. It is a very strong audio platform that also streams Bluetooth from your iPod or your phone. It goes into headsets, sound bars, speaker docks and then the next big opportunity is to have the same system streamed into the entire house using a combination of Bluetooth and Wi-Fi.

Why should people get into designing devices with Bluetooth Smart?

With Bluetooth Smart, we talk about wearables such as the Nike+ FuelBand, and an enormous amount of verticals, that are still small. But a lot of people are very excited about it because it can grow exponentially. Voice, music and automotive might be the sectors which pay the bills today for us. However, Bluetooth Smart is very exciting but it is still more investment than revenue, although we have shown very good growth. People talk about a 100 billion connected devices in 10 years' time, a large chunk of these are going to be connected through Bluetooth Smart.

What is the most exciting product that you have seen in audio?

The excitement is all about streaming audio. Two products were launched at CES this year, which are pieces of silicon and software that allow you to access any musical source that you have. You can take it from the cloud and stream to Spotify, or you can take it from the DLNA server, iPod and Android. The stream gets into your hub, and then your hub can distribute it to your entire home with even different streams to each room. This is really exciting and there exists a phenomenal growth opportunity in the voice and music arena for the coming years.

When can we see mesh networking in Bluetooth Smart?

In CES, we demonstrated which I thought was a world first, a design that functioned through Bluetooth Smart meshed. In our booth, we had 25 different lamps that you could actually address individually through what was a

long hallway. At the far end of the hallway, beyond normal Bluetooth Smart range, were these lights that needed to be controlled. But because it was meshed, the control was not a problem. We could address individual lights, a group of lights, or all of them at once.

What would it take to do something similar with Zigbee?

What I feel is that you can go Zigbee and get the mesh capability, but now you need a Wi-Fi access point, and it becomes cumbersome starting there. For instance, in Philips products that are currently available in the market using Zigbee, there are only three lamps and you need a separate access point for it. So, if you have 21 lights you then have seven different queues on your phone, and you can see how it goes on from there.

What is interesting about designing for automotive industry?

Automotive market as a whole has a little bit more visibility, and the reason is that it is the very nature of that industry. Design cycles are very long and it can take you up to two years to start the design process, ending it and actually getting the chip into the car. Now, once your component is in the car, it is not easy to get it designed out. So although it takes a while to get in, the benefit is that once you are in, you stay in. Cycles are longer when compared to consumer electronics where it is mostly six months, and therefore the visibility is a little bit better. It is very competitive, just as competitive as the smartphone which is ultra competitive.

How does the design process usually work in the automotive industry?

We talk to the OEMs to be able to understand what it is that they want, so that we can put it on our roadmap. The actual sale is usually to so-called tier 1 companies. These are companies that are selling a complete sub-system, like a head unit or a powertrain or what have you, to the car manufacturer. For instance, if you get a win with Toyota, you are going to deal with a company called Denzel. Or there is another company called Continental, or you have Visteon, or Robert Bosch. These are the tier 1 customers that we physically ship to, where they integrate it into a larger subsystem and that subsystem gets installed into the car. In automotive infotainment we are moving from a world in which we sell individual components into a completed infotainment bundle, using SoCs for Bluetooth, Wi-Fi, etc. ●